Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI3461-WE01

Title: MEng and BEng Engineering

Advanced Computer Systems and Digital Electronics 3

Time Allowed:	2 hours	
Additional Material provided:	None.	
Materials Permitted:	None.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries:	YES	

Instructions to Candidates:	Answer ALL questions. Use a separate answer book for each section. All relevant workings must be shown.

Revision:	

Section A
Q1

a) What is processor instruction pipelining and how does it improve performance?

[20%]

b) RISC is sometimes described as “load/store” architecture. What does this mean and how does it improve performance?

[20%]

c) EITHER

Explain, with a simple code example, how branching affects pipeline performance and what can be done about it.

OR

Explain how Harvard architecture affects pipeline performance; and why different parts of a Harvard memory architecture might employ different cache policies.

[60%]

Q2

a) Explain what is meant by “locality of reference” and why it is important for memory caches.

[12%]

b) Give one reason (each) why locality of reference occurs within

- Instruction fetch
- Data load

[32%]

continued

c) Find a series of 10 memory requests, containing no more than three different address locations, which will have a hit rate of 0% with a direct-mapped cache. Justify your answer, including the cache line length and cache size you are assuming (both should be greater than 0).

[20%]

d) Explain what status bit(s) would be required for each cache line, and how they would be used, to implement ONE of the following

- Write-back cache
- Least-recently-used line replacement in a 2-way set-associative cache
- A cache coherency protocol for a multi-core processor

[36%]

Section B

Q3

a) Explain the role of a configurable logic block (CLB) in an FPGA, and describe how a simple CLB might have been implemented.

[12%]

b) Consider the following two Boolean functions, given in sum-of-minterms form. Inputs of the function are named a, b, c, d. Implement these two functions on a PAL section with five inputs and two outputs, where each output OR gate of the PAL has three AND gates as inputs. Draw the corresponding circuit diagram.

$$F_1(a,b,c,d) = \Sigma m(2,12,13)$$

$$F_2(a,b,c,d) = \Sigma m(1,2,8,12,13)$$

[24%]

continued

c) Figure Q3c shows a finite state machine for a simple synchronous 2-bit binary up counter. ‘CNT’ is the 2-bit output of the counter. ‘reset’ is an asynchronous input, to reset the counter output to “00” when reset=1. Use an additional system input named ‘Up’ to allow the counter to count up when Up=1 and count down when Up=0 and modify the state machine accordingly. Write VHDL code for the modified state machine.

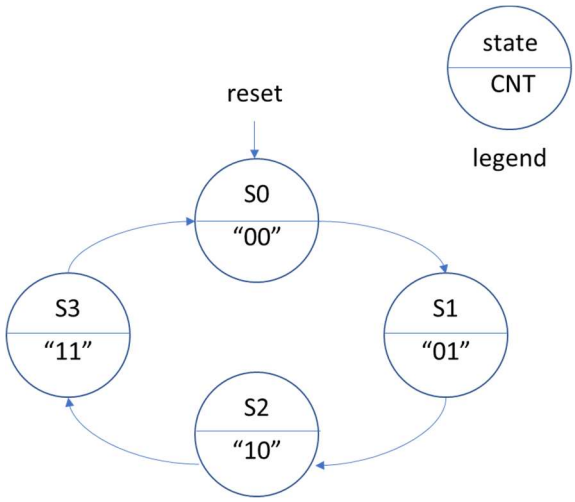


Figure Q3c

[32%]

d) Explain how a 4×4-bit binary multiplication can be implemented using a ripple-carry multiplier. Draw a figure and show how the partial products are summed using full adders in this configuration. Then explain how the ripple-carry multiplier compares with carry-save multiplication in terms of speed.

[32%]

continued

Q4

a) i) Explain what is meant by the following terms in the context of digital logic fault testing:

- test vector
- stuck-at-1 fault
- test vector coverage
- boundary scan

ii) For the following Boolean logic expression, draw the corresponding circuit and find a minimum set of test vectors that will test all single stuck-at-1 faults. For each test vector, specify which faults are tested.

$$Z = (A+B+C).(D+E+F).G$$

[28%]

b) An iterative logic network is required which can produce a high output when a pattern of alternating of 1s and 0s in an 8 bit word is detected (i.e. 01010101 or 10101010). For all other cases the output of the network is low.

Design an iterative logic network to meet this specification. In your answer, you should clearly outline each stage of the design process. You do not need to draw circuit diagrams but should provide the simplified Boolean logic expressions that are required to implement the basic cells and any special adjustments for the first and last cells in the network.

[44%]

c) Compare and contrast the properties of iterative logic networks with asynchronous sequential logic networks.

[28%]

END